| Problem Chosen | 2020 MCM/ICM Summary Sheet | Team Control Number |
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# T-DNA: Team Promotion Based on Double Network Analysis and Assessment


**Summary**

Faced with increasing chanllenges, we rely on teamwork to address the most challenging problem. A team needs goods modes of assembling, interactions and leadership to achieve success. How can we assess these factors and promote team success? Modeling and analysing the datas in competitive team sports can be instructive. In this paper, we model the ball passing events of 38 soccer matches related to team Huskie with **double network analysis(T-DNA)** and gain some insights into the problem.

In the first section, we build a **simple weighted network** to model the passes in a single game and in the whole season. We analyse the **degree distributions** of the networks and find the Huskies put emphasis on defence and a star player. We introduce **motifs** to depict their tactic on a micro scale and find one of the feature of team Huskie is **asymmetry**. There is often a leader in the tactical unit while in other teams the tactical unit is symmetric.

To further analyse the assembling and leadership of the soccer team, we derive a **Reciprocal Network** from the first model. We extract the **shortest paths** of the new network with Dijkstra algorithm and Floyd-Wawashall algorithm. Based on the shortest paths matrix, we define the **contribution coefficient** to quantify the contributions of teamates. We find the most important player in Huskie contribute 7 times more than the average to the pass network of the team.

In the third part, we apply **Entropy Method** to comprehensively consider performance indicators that reflect successful teamwork and test our result with the actual results of the matches. With the properties and indicators we found in our model, The comprehensive accuracy of our prediction reach 73.6%.

After testing our model, we give the coach some **advise** based on the insights gained from our T-DNA model. More importantly, we further consider the links and differences between soccer games and other teamworks to **generalize** our result. We suggest that the most important factor is the common **goal** of the team.

We analyze the sensitivity of the model mainly by checking the result that T-DNA produces on a subset of the dataset. We develop an algorithm to carefully select the ball pass data but find the main feature vary within 9% in general, which means our model is not sensitive to small changes in the given data. We also give explanation of the sensitive part of our model.

We conclude our work then.



**Key Words:** team network, motifs, shortes path algorithm, principal component analysis.

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# 1 Introduction

## 1.1 Problem Background

Team success is much more than the sum of individuals. It is more powerful, but also more complicated. As we often work in teams to solve challenging problems, it is important for us to study what makes a team successful.

Competitive team sports provide plenty of data for us to explore team processes. Here we get the data of a soccer team named Huskie throughout last season and want to analyse how their **interactions impact success** and to provide **better strategies**.

We also apply such analysis and strategy to other teamwork besides sports to improve efficiency.

## 1.2 Problems Restatements

Quantify and formalize the structural and dynamical features which are related to successful teamwork. Based on previous work, help identify specific strategies to improve teamwork and give a generalization.

To be more specific, we need to achieve the following steps:

1. Create a ball-passing network. Then identify network pattern, extract structural indicators and describe network properties.
2. Indentify indicators that reflect successful teamwork.
3. Inform the coach about structural strength and weakness of the team. Give advice about structural improvement based on network analysis.
4. Consider more factors and generalize the model to design more effective teams in other aspects.

Here we need some clarification. In this paper, we interprete some statements as follows :

- network property: Description of features of a network, such as connectivity, degree distribution.
- structural indicators: The variables that describe network properties.
- network pattern: Network property that are directly related to the stratege.
- Indicators that reflect **successful teamwork**: In this paper, points and wins is the **basic standard** of successful teamwork. Thus the indicators should not contain points or wins since it is trivial. Good indicators should be controlable and have high relativity to the result of the game.

### 1.3 Our Work

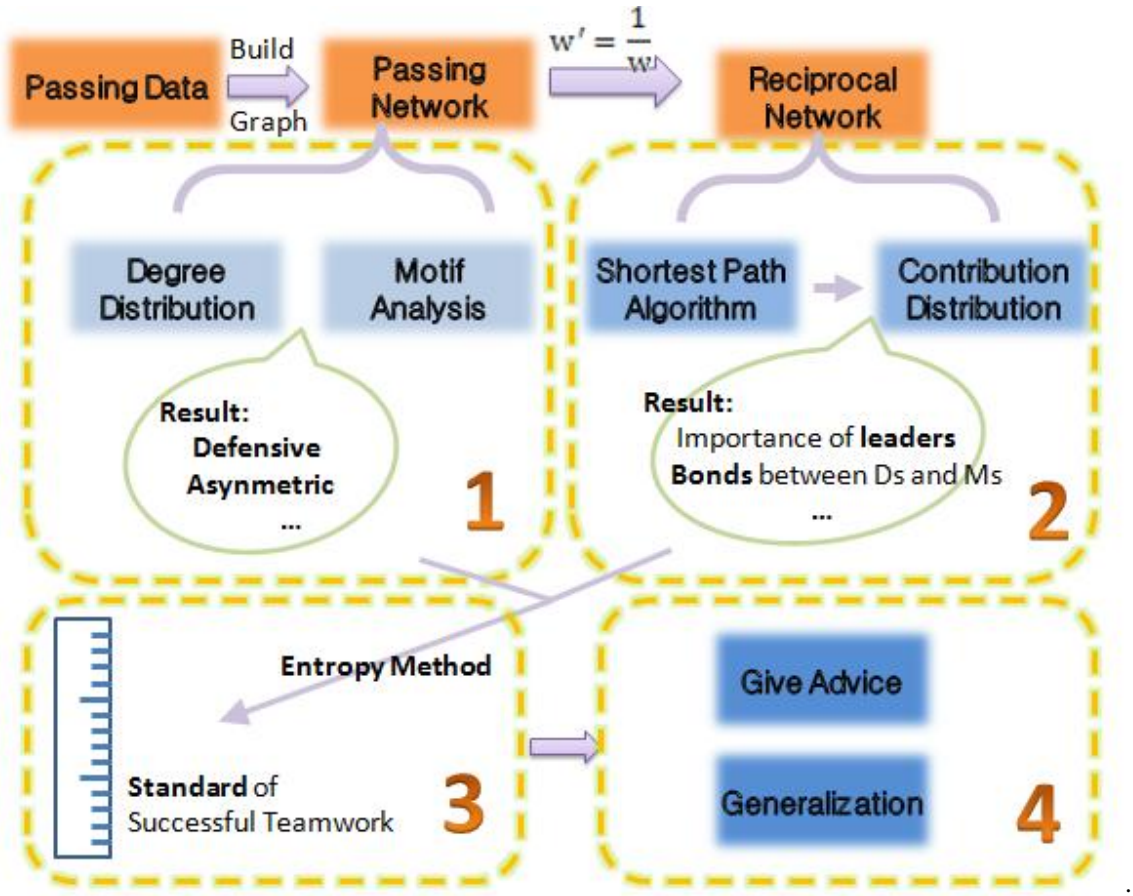



Figure 1: The Structure of Our Model.First we build a simple network with the passing data of the matches related to team Huskie. We analyse the **degree distribution** and **motifs** in this model and find the Huskies put emphasis on defence and a strong midfielder. Then we develop $ReciprocalNetwork$ and apply shortest path algorithms to find the **customary route** and **contribution distribution** of the team. To mix the indicators together and test the effect of our model, we apply **Entropy Method** to weigh the network indicators and provide a standard of successful teamwork. After testing the effects of the standard with the datas, we provide advise and generalize our discovery based on it.

## 2 Assumptions and Notations

### 2.1 Assumptions

To simplify our problems, we make the following basic assumptions and explain their rationales, each of which is properly justified.

**Assumption 1.** ***All players are rational, they pass the ball for vectory.***

*We assume all players pass the ball to one of their teammates not because they are good friends, but to seek chance for goal. It is reasonable to believe those prefessional players would do so.*

**Assumption 2.** ***The average individual ability is similar among different teams.*** *We assume the result of the game mainly depend on the team network rather than individual ability, since cooperation is more important in soccer games and the players are trained in the same level.*

**Assumption 3.** ***Forwards focus on offense while defenders put emphasis defense.***

We assume the strategy of the team is not too weird to understand. We analyse the team formations and the feature of strategy based on such common sense.

### 2.2 Notations

The primary notations used in this paper are listed in **Table 1**.

Table 1: Notations

| Symbol | Varible Name&Explanation |
|---|---|
| $v_{xx}$ | Player vertex, to denote the player xx$\in$ {g1,m1,d1,f1,$\cdots$} |
| $e_{xx-xx}$ | Passing edge, to denote the passing between players. |
| $w_{xx-xx}$ | Edge Weight, the weight of $e_{xx-xx}$. |
| $Z$ | Z score, to indicate the significance of a motif |
| $k_{clustering}$ | Clustering Coefficient |
| $S_{motif}$ | Score of Motifs |
| $k_{contribution}$ | Contribution Coefficient t |
| $\sigma^2_{contribution}$ | Contribution Variance , the variance of contribution distribution |
| $\sigma^2_{degree}$ | Degree Variance, the variance of degree distribution |
| $\boldsymbol{W_{shotest}}$ | Shotest Distance Matrix |

### 2.3 Terminologies

- Graph: $G = (V, E)$, where V is the set of vertexes, E is the set of the Edges.
- Subgraph: $G' = (V', E')$, $G'$ is a subgraph of $G$ if and only if $V' \subseteq V$ and $E' \subseteq E$
- Induced Subgraph: $G' = (V', E')$ is a subgraph of $G = (V, E)$, $v_i \in V$, $e_{ij} \in E$ is a edges linked to $v_i$. $G'$ is a induced graph of $G$ if and only if for $\forall i, j, v_i \in V' \Rightarrow e_{ij} \in E'$.

## 3 Simple Network Model

In this section, we build a weighted network model for the ball passing between players. To analysis the micro structure of the network, we extract and classify its **motifs**. To explore the macro structure, we define node weight by Floyd-Warshall algorithm and find the relationship between the node weight distribution and the result of the games.

In this section, we further need the following assumptions:

**Local Assumption 1.** ***All passes are equivalent when it comes to network properties.***

*We focus on the topological structure of network in this section and neglect the time, position differences between passes. We will analyse such factors in "Sensitivity Analysis" to justify this assumption.*

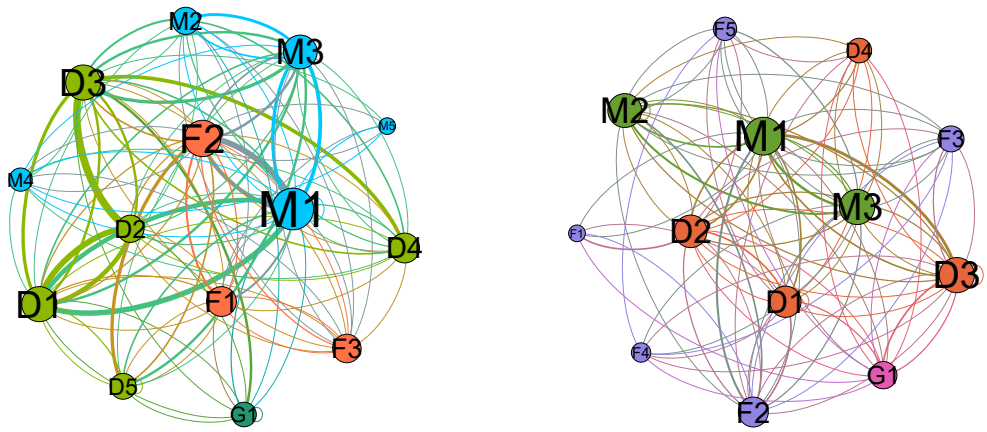
### 3.1 Build the Network

Soccer is a game of cooperation. The players in the champion team may not be the most talented. But they must be well-coordinated. To describe how well the team cooperate and assess the efficiency of their strategies, a reasonble way is to look at the passing events between them. Thus we introduce some methods of graphy theory to analyze the passing events.

Denote each player of the Huskies as a vertex $v_{xx}, xx = g1, d1, d2, m1, m2, f1, f2, \cdots$, players in the opponent teams as $v_{xx'}$. For example, $v_{g1}, v_{f1'}$ represents the goalkeeper of the Huskies and forward1 of the opponent team respectively.

If there are passing events between the two players, we add a edge $e_{xx-xx}$ between them. Define the weight of the edge as the number of passing events between them (in a half, game or season, depending on context), denoted as $w_{xx-xx}$. For example, defender1 of the Hussikies passes to forward2 of the Hussikies twice in a game, then $w_{d1-f2} = 2$ in the network of this game.

Here we pick the network of a typical match: .




(a)Network of the Huskies. (b)Network of Opponent1.

Figure 2: The sizes of the nodes indicate the total degree of the vertexes. The large the size is, the more the player passes the ball in the game.

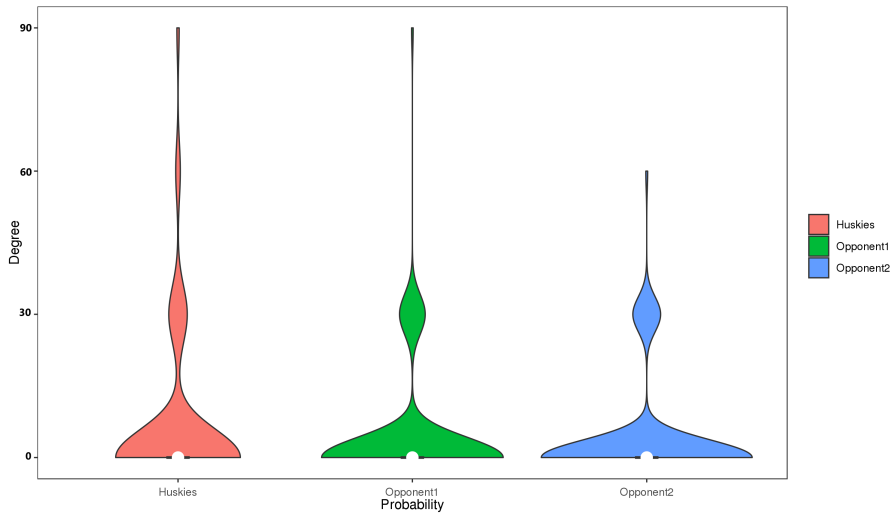
We can find some intresting facts from the graph above:

- Strong M1: At a glance of Network(b), we will notice M1, the largest node(if we plot the two graph together with the same scale, it will be more obvious). It means that Midfielder1 in the Husskies passes the most balls in the match, while the other 3 midfielders have less chance of passing the ball. By comparison, the 3 midfielders in the oppinent team are almost equivalent. We can infer that M1 is a **star player**, causing some **imbalance** in the team.
- The strong bonds between the three defenders: We can also see the bold line between D1, D2, and D3. It shows the strategy of the Huskies are **defensive** rather than offensive. The

main links in its opponent's network are among the midfielders, which also verify our deduction.

## 3.2 Degree Distribution

A graph with more than ten vertexes is still too complex for us to analyse just by intuition. To extract more information from the data. We calculate the degree distribution of the Huskies in the matches of the whole season.




The Average Degree Distribution of the Huskies Passing Network in a Season.

Compared with opponent2, the top end of the violin graph represent the star midfielder of the Huskie.

Compared with opponent1, the middle part of violin is a little thinner, indicating the asymmetry of the team to some extend.

We can see the violin base of the Huskies are thinner than that of other team. It shows that the Huskies pass ball more than the other players, which reflect their **flexibility**.

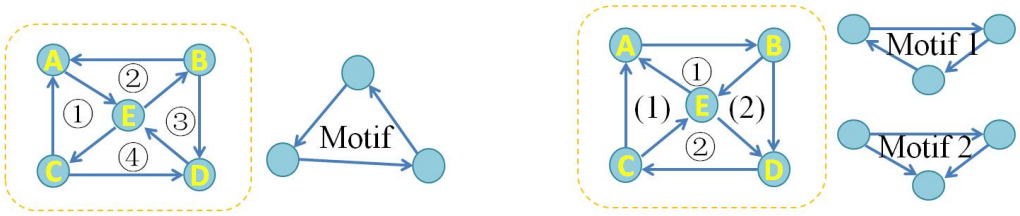
## 3.3 Motifs Analysis

Intuitive analysis help us grasp the general feature of the passing network. But how to describe the similarities and the differences between these networks more precisely?

### 3.3.1 Introduction of Motifs

We get inspiration from several papers related to motifs. Motifs of a graph $G$ are the subgraphs that appears more frequently in $G$ than in other networks similar to $G$.

In the two examples above, we can tell the motifs by intuition. But for general cases, we need methematical description of "more frequent" and "similar".




Example(a):Network with only one motif. Example(b):Network with two motifs.

Figure 3: We can see Example(a) and Example(b) are similar, but there are also some differences. In Example(a), three-vertex loop appears 4 times and it is the only three-vertex induced subgraph of (a). In Example(b), there are two kinds of three-vertex subgraph, both of which appear twice. The difference between the motifs and the frequency they appear in the graph depicts the structure difference of networks on micro scale.

### 3.3.2 Definition and Algorithm of Motif

According to [7], we define similar graphs of $G$ as the graph with the same in and out degree distributions. We denote such graphs as $G_1, G_2, \cdots$. The pattern $G'$ appears $N_{real}$ times in $G$, $N_1, N_2, N_3, \cdots$ times in $G_1, G_2, G_3, \cdots$. We assess how frequent the pattern appear in $G$ by looking at how much $N_{real}$ surpass the average of $N_1, N_2, N_3, \cdots$.

We define $Z$ score:

$$Z = \frac{N_{real} - \mu_{rand}}{\sigma_{rand}} \tag{1}$$

where

$$\mu_{rand} = \frac{\sum_{i=1}^{n} N_i}{n}$$

$$\sigma_{rand} = \sqrt{\frac{\sum_{i=1}^{n} (N_i - \mu_{rand})}{n}}$$

are the average and standard deviation of $N_1, N_2, N_3, \cdots$

Now we can give the formal definition of motif.

**Definition 1.** *A graph $G'$ is called a motif of graph $G$ if*

- *$G'$ is a **subgraph** of $G$.*
- *$G'$ is **(weakly) connected**.*
- *Let $\mathscr{G} = \{G_i | G_i$ has **the same in and out degree distributions**,i=1,2,3,$\cdots$ with $G\}$,*
  *$G'$ appears $N_{real}$ times in $G$, $N_i$ times in $G_i$.*
  *Let $\mu_{rand}$ and $\sigma_{rand}$ be the average and standard deviation of $\{N_i | i = 1, 2, 3, \cdots\}$.*
  *The Z score: $Z = \frac{N_{real} - \mu_{rand}}{\sigma_{rand}} > 1$*

Here we choose 1 as the threshold value of Z score after we explore the Z scores in different passing networks.

With the definition above, we can derive an algorithm to determine the Z scores of motifs.

We details the algorithm from [7] as follows:

**Algorithm 1:** Find the Three-Vertex Motifs

**Input:** A graph $G$
**Output:** The three-vertex motifs of $G$ (In the order of Z score)

```
𝒢 = The list of some randomly generated graphs has the same in and out degree
    distributions with G
ℳ = The list of all 13 three-vertex motifs.
motifs=An empty list.
for pattern in ℳ do
    N_real = The time pattern appears in G.
    for G_rand in 𝒢 do
        N_i=The time pattern appears in G_rand
    μ_rand = the average of {N_i | i = 1, 2, 3, ···}
    σ_rand = the standard deviation of {N_i | i = 1, 2, 3, ···}
    if (N_real − μ_rand)/σ_rand > 0.5 then
        Put pattern in the list motifs.
Sort the elements in list motifs.
output motifs
```

The 13 motifs we study in this paper are listed below. [7]

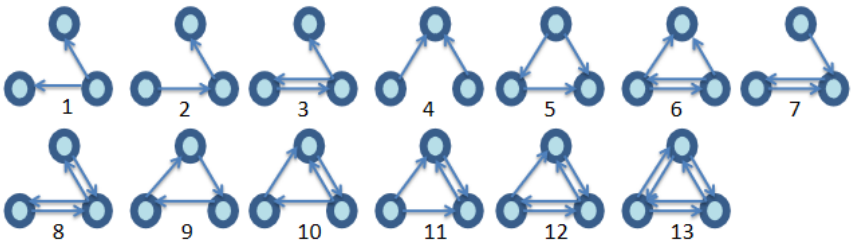



Figure 4: All the three-vertex motif.

In soccer game or other teamwork with small amout of people, there is connection in every two members. Thus we can infer that 1, 2, 3, 4, 7, 8 is usually not the motif of a real team network.

### 3.3.3 Analyse the Result

The motif preference of the Huskies are quite unique.

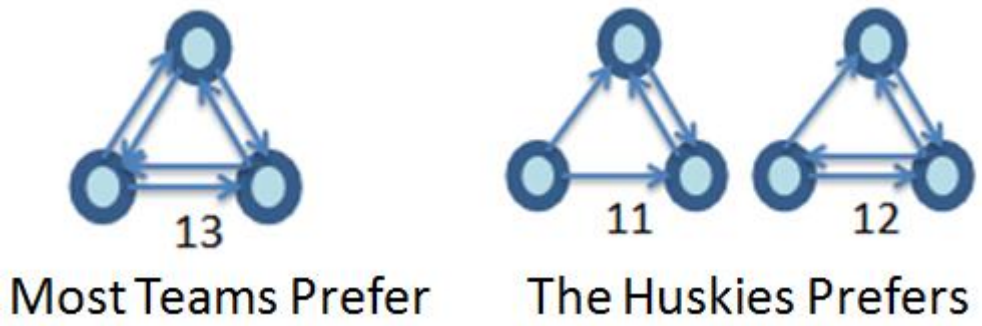
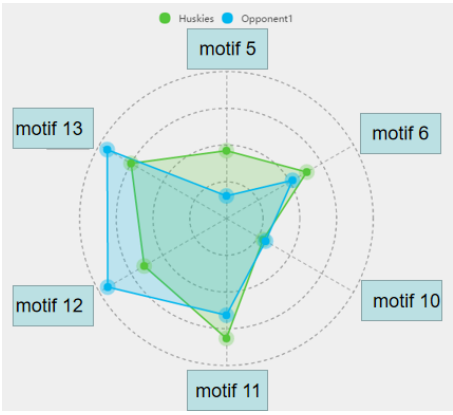
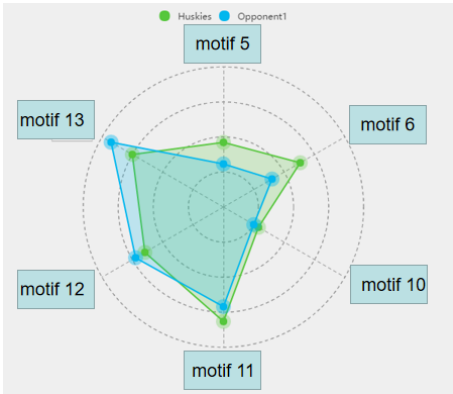



Figure 5: The Comparison of Motif Preference

The preference of motifs reflect the tactical thinking of a team on a micro scale. Most team choose motif13 since it is a **fully connected** graph. The players in a tatic unit are all equivalent and communicate with each members without bias.

The Huskie tend to choose a leader in the tactic unit, which cause the asymmetry in the team. It may help them well organized, but also restricts their cooperations.




Radar Map of the Huskies' Motifs(compared with their opponents). It shows that the Huskies try a variety of motifs. Such **diversity** in tatic compensate for the defect that their network are less connective in micro scale.

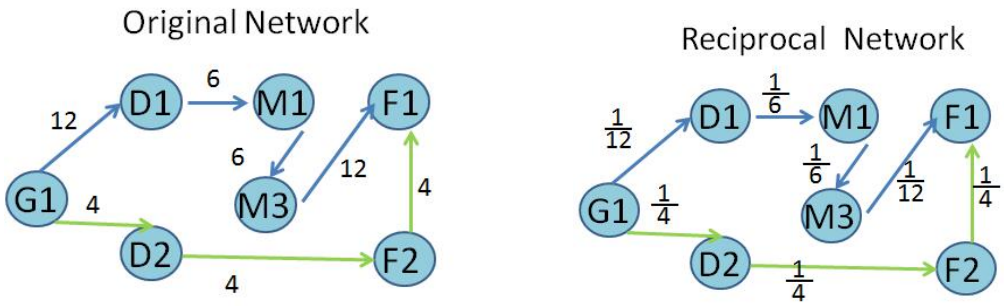
# 4 Reciprocal Network

Inspired by dual circuit, we define the weights of the passing network as the reciprocal of that in the previous section. It is also simple. But we find it useful as we apply shortest path algorithms to it and obtain meaningful result of the network properties.

## 4.1 Customary Routes

In the last section we explored the network structure on micro scale (motifs) and find the local tactic frequently used by Huskies and other teams. Now we want to know the global features and strategy that the passing network implies.

One of the important global feature is the **customary route**. That is, how do one of the Huskies usually passes ball to the others. For example, if ball can be passed from G1 to F2 by G1-D1-M1-M3-F1 or G1-D2-F2-F1. Which way shows their habits and indicates their stategy more? The latter one needs less intermedia players while the weights on the former one are larger, which means the members on this route tend to pass between each other.




(a) (a)

Figure 6: An Example of the Idea: In the left graph(a), we can not tell which route from G1 to F1 is more customary for the team. The green one passes less vertexes while the blue one has larger weight (passing frequency). But in the reciprocal network, we can simply add the weight up to compare. Since $\frac{1}{4}+\frac{1}{4}+\frac{1}{4} > \frac{1}{12}+\frac{1}{6}+\frac{1}{6}+\frac{1}{12}$, the blue route is more customary.

To describe and compare the combination of the two factors, we keep the topological structure unchanged but take the **reciprocal** of the weights of edges to derive a new network $G_r$. Similar to the definition of **dual circuit**. We give a formal definition as follows.

**Definition 2.** *Let weighted graph $G = (v_i, e_i, w_i)$. We call graph $G_r = (v_i, e_i, w_i')$ the reciprocal network of $G$ if and only if $w_i' = \frac{1}{w_i}$ holds for all $i$.*

**Definition 3.** *The customary route from vertex A to vertex B is the shortest path from A to B in the reciprocal network.*

To get the customary route between two specific point, we apply **Dijkstra Algorithm**. The complexcity is O($nlog(n)$).

To get the length of the customary routes between every two vertexes, we apply **Floyd-Wawarshall Algorithm**. [1] The complexcity is O($n^3$). As the number of vertexes are limited, there is no problem.

By Floyd-Wawarshall Algorithm, we can get a shortest distance matrix, denoted as $\boldsymbol{W_{shotest}}$. We visulize $\boldsymbol{W_{shotest}}$ of the Husskies in the first match.

We can see from the heatmap that the upper-left corner is quit bright, indicating the **close link between the defenders and the forward**. Such feature makes the offense of the Huskies quite efficient. We know from the simple network that the defenders and the forwards are not usually connected directly. Thus the heatmap actually show that the midfields plays an important role in their **quick attack** tactic.

Consider the fact that it is a defensive team. It is not difficult for us to understand that the efficiency to pass the ball to the front is critical for the Hukies to win.

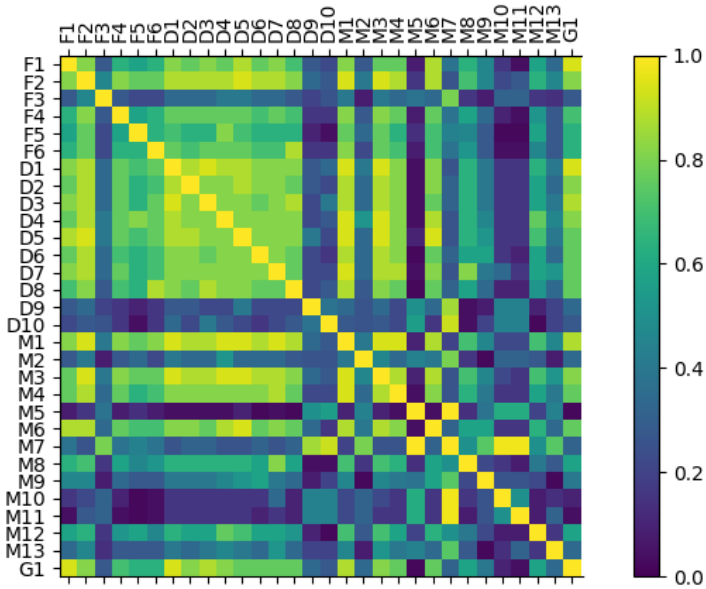



Figure 7: Heatmap of the length of all customed paths.

## 4.2 Contribution Analysis

To analyse the global feature of the team and describe their strategy in graph language. We also need to weight the importance of each vertex. Based on our global assumption1, all the players are rational. Thus the more playerA passes ball to playerB, the more important the edge $e_a b$ is in the team strategy.

In the simple network model, we describe the importance of a vertex by its in and out degrees. It is useful, but does not derectly relate to the player's contribution . To produce a more convincing indicator, we can extrat something in customary route.

A player in the center of the network helps to pass many balls between two members far away. While a player at the corner seldom helps to build the connection between other teammate.

To convey such meaning in the indicator, we intepreté the concept "contribution" as how much a player **contribute to the connectivity**.

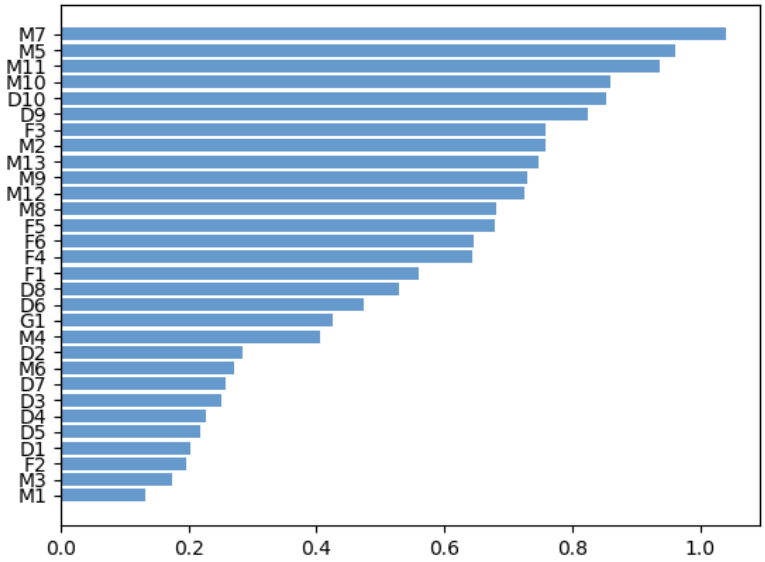
To quantify this factor, we calculate the average of the propotion that a vertex is in the shortest path of any other two vertexes and give the following definition:

**Definition 4.** *Let $\boldsymbol{W_{shotest}}$ be the $n \times n$ shortest distance matrix ,$w_{i,j}$ is the length of the shortest path from $i$ to $j$. Then the contribution factor of $k$ is*

$$k_{contribution} = \frac{2}{n(n-1)} \sum_{1 \leqslant i \leqslant j \leqslant n} \frac{w_{i,j}}{w_{i,k} + w_{k,j}} \tag{2}$$

We can see from the definition that $0 < k_{contribution} \leqslant 1$. $k_{contribution} = 1$ if and only if all the shortest paths in the graph pass the vertex $v_k$. A good example is that in a five-vertex graph that forms a cross, that the $k_{contribution}$ of the center vertex equals to one.

We can see that the contribution varies much from players to players. The most important




Average $k_{contribution}$ of the Huskies member all over the season.

player contribute 7 times more than the least one according to this indicator.

By comparison with other teams, we find the variance of the contribution distribution of the Huskies is $3.08$ times of the average level. Which again describe the **asymmetry and imbalance** of the team formation.

# 5 Entropy Method Model

Through the above model analysis, we deeply study the team's tactical and cooperative attributes and properties, and have some quantitative description. In order to better find the factors that determine the success of teamwork, we need to **use a model to evaluate the overall performance of a team** by combining the data of various factors in the team during the game, and **find the main factors for analysis**.

## 5.1 Introduction of Entropy Method

According to [8], Shannon and Weaver proposed the entropy concept, which is a measure of uncertainty in information formulated in terms of probability theory. The greater the entropy it contains, the less uncertainty, entropy and information it contains. Shannon's concept is capable of being deployed as a weighting calculation method. Entropy weight method is an objective weighting method because it only depends on the discreteness of the data itself.

Through the analysis of the network model, we have a lot of attributes and data about the passing cooperation of teams. We also have abundant statistics from the games, and that have a lot to do with successful teamwork and winning, like total shots and total save attempts. What is more, we consider the Huskies' performance is related to the strength of the opponent, so we need to consider the impact of the opponent's defense and offense on the Huskies.

Given that there are many factors that affect a team's performance in the game, we want to use these factors to **find out the main indicators that determine the team's performance**. Considering the characteristics of entropy, we think that entropy method can help us effectively evaluate the team's performance in the game, and **the factor weight** can indicate the importance of this factor to the team cooperation.

Therefore, we will evaluate the teamwork performance through the entropy method. We can **verify the accuracy of this model by the game results of the Huskies and their opponents**.

## 5.2 Determine the Factors

Before using the model we need to **identify the factors that affect the game**. In general, teams in the game focus mainly on offense and defense. According to the available statistics, we think the attack mainly from the passing and shotting, while the defense mainly from the goalkeeper's save. We also want to analyze the adaptability of the team. In this respect, we mainly consider whether the Huskies will be greatly affected by opponents' fierce attack and defense, that is, the impact of the opponent on the Huskies' contribution to the evaluation model.

Then we discuss **the factors** we choose to describe team's performance.

### 5.2.1 The Passing Factor

In the first model, we analyzed the team's passing network and obtained the score of Motifs, which describes the team's common coordination tactics. This parameter can **effectively show the quality of the team's passing in key offense and tactics**. So we use **the score of Motifs** as the first factor $x_1$ to describe team passing effect.

### 5.2.2 The Shotting Factor

The Huskies' shotting situation is not fully considered in the passing network, but it is obviously **an important manifestation of the offensive effect**. Huskies' shotting is the ultimate goal of effective passing coordination. The data that can best reflect the effect of team Huskies' shotting is the number of shots in the game, so we take **the number of shots** as the factor $x_2$ in the evaluation model.

### 5.2.3 The Defending Factor

The team's defensive tactics also need close coordination and response, so defense is also an important factor in evaluating the effect of teamwork. A good defensive tactic often ends with a successful save or steal, so **the number of saves and steals** a team makes in the game is an intuitive reflection of the team's defensive quality.

The original data does not provide team's steals, but we can **dig out the team's steals from the passing data**: we find that the pass of the passer and receiver belonging to different teams is the steal. Since the number of steals and saves describe the team's defensive effect, in order to avoid too many factors, we take **the harmonic average of them** as the parameters to describe the team's defensive effect.

$$x_3 = \frac{1}{\frac{1}{\textit{number of steals}} + \frac{1}{\textit{number of saves}}}$$

### 5.2.4 The Opponent Factor

We also need to consider the influence at the team level, mainly the adaptability and resilience of the team. The adaptability of the team is mainly influenced by **the opponents and the external environment**. In a match, a threatening attack or a brilliant save from an opponent will have an impact on the team, and if the team is not adaptable, it will lose the initiative of the game. So the threat of opponent can be described by **the number of opponent's shots and saves**.

We also take other factors into account, such as home or away, match weather, and player morale. But the impact of these factors is not significant compared with the impact of the opponent, and these factors are difficult to quantify, so we do not consider these off-field factors.

For the sake of the simplicity of the calculation and results, we take **the harmonic average of the number of shots and saves** of the opponents as an evaluation factor.

$$x_4 = \frac{1}{\frac{1}{\textit{number of opponent's shots}} + \frac{1}{\textit{number of opponent's saves}}}$$

## 5.3 EM Model Algorithm

Let factors for entropy method be $x_1, x_2, \cdots, x_m$, and we have $n$ evaluation objectives (team performances). And the $ith$ evaluation objective's $jth$ is $a_{ij}$.

1 For the precision of evaluation, we **standardize index value** from $a_{ij}$ to $\tilde{a}_{ij}$. If $a_{ij}$ is an possitive indicater, we have:

$$\tilde{a}_{ij} = \frac{a_{ij} - \min\{x_1, x_2, \cdots, x_{nj}\}}{\max\{x_1, x_2, \cdots, x_{nj}\} - \min\{x_1, x_2, \cdots, x_{nj}\}}, i = 1, 2, \cdots, m, j = 1, 2, \cdots, n$$

If $a_{ij}$ is a negative indicater, we have:

$$\tilde{a}_{ij} = \frac{\max\{x_1, x_2, \cdots, x_{nj}\} - a_{ij}}{\max\{x_1, x_2, \cdots, x_{nj}\} - \min\{x_1, x_2, \cdots, x_{nj}\}}, i = 1, 2, \cdots, m, j = 1, 2, \cdots, n$$

2 Calculate **the proportion** of the $ith$ object in the $jth$ index:

$$p_{ij} = \frac{\tilde{a}_{ij}}{\sum_{i=1}^{n} \tilde{a}_{ij}}$$

3 Calculate **the entropy** of the $jth$ index:

$$e_j = -\frac{1}{ln(n)} \sum_{i=1}^{n} p_{ij} ln(p_{ij}), j = 1, \cdots, m$$

4 Calculate **the weight of each index**:

$$w_j = \frac{1 - e_j}{\sum_{j=1}^{m} 1 - e_j}$$

5 Use weight to calculate **the score of each team**:

$$s_i = \sum_{j=1}^{m} w_j \tilde{a}_{ij}$$

## 5.4 Verification of Our Model

We sift through the data of the Huskies and calculate the scores for the Huskies and the opponents. We also give the Huskies' league points against different opponents (3 points for a win, 1 point for a draw, and 0 point for a loss) from the actual data.

Table 2: Caculation results and verification

| teamID | O1 | O2 | O3 | O4 | O5 | $\cdots$ | O18 | O19 |
|---|---|---|---|---|---|---|---|---|
| Socre of Huskies | 55.35 | 46.04 | 49.32 | 40.58 | 43.78 | $\cdots$ | 47.70 | 54.50 |
| Score of the opponent | 36.51 | 48.13 | 54.71 | 54.85 | 51.58 | $\cdots$ | 40.08 | 34.98 |
| League points Huskies gets | 6 | 1 | 1 | 0 | 0 | $\cdots$ | 4 | 2 |

According to the data, we find that the team score can effectively reflect the teamwork performance in the game. From $table2$, we can see Huskies scores more points when its score is higher than their opponents. If the Huskies scores less than their opponents, they have a significant chance of losing the game. Therefore, we **verified the effectiveness of the EM Model**.

## 5.5 Contribution Analysis

We calculate **contributions rates of different factors**:

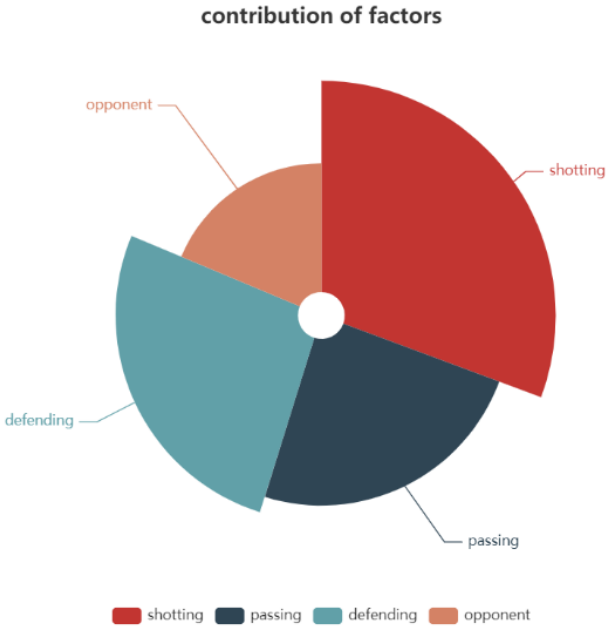



Figure 8: In the figure, we find that the contribution rate of Huskies shotting was up to $30.67\%$, which is the most important factor. But we find that the passing factor and the defense factor also have a high weight, indicating that offense and defense were almost equally important. The influence of the opponent reaches $18.69\%$, showing that the influence of the opponent is not negligible

We can see the Huskies team performance is mainly determined by **the overall offense**, which consists of a combination of passing and shotting, which accounts for $57\%$. At the same time, Huskies team performance is likely to be affected by the opponent, that is to say, they are easy to be targeted and contained by the opponent. if opponents use strategies to restrain Huskies' play, Huskies are easy to be passive.

# 6 Advice for the Huskies

Based on the insights gained from our teamwork model, we find the efficient strategies that has been adopted by Huskies. We also propose several improvements according to network analysis.

## 6.1 Efficient Strategies they adopted

- **Defensive formation**: Huskies adopts a more defensive formation when it faces offensive opponents. They change to a classic 5-4-1 formation, more focused on defense, to avoid being more passive.
- Counter-attacks: Figure 6 shows the defenders have strong connection to forward, indicating the defenders tend to deliver the ball to forward by long pass to counter-attack. Counter-attacks and defensive formation are two complementary strategies. Defensive formation stops Huskies' opponents from scoring, once our defenders steal and give it to forward by long pass. Opponents will be unprepared and Huskies gets chance to score.

## 6.2 Improvements

- **Balance the weight of offense and defense.**: Our research above shows that the Huskies is into defensive formation and have fewer trial to attack actively. Since attacking is the only way to score, Huskies need to strength attacking abilities.
- Balance the side: Section 3.1 indicates that Huskies is not balanced in the midfield. They got single important midfield player, so they have to launch an attack on one side. That will be observed by opponents and attracts purposeful defense. A practical proposal is to sign in a good midfield player.

# 7 Generalization

## 7.1 Design Effective Teams

Soccer is a typical team sport, so we can generalize these methods to a higher level.

### 7.1.1 Strengthen communication within the team

In the soccer network, we find that the Huskies are **not closely connected in key tactics**, which leads to the ineffective execution of tactical coordination.

In a team, if the nodes are not connected enough, network information traffic will decrease and resource utilization will decline. Therefore, the harmonious relationship and sufficient communication in the team can **reduce the estrangement of cooperation and make the team more effective**.

### 7.1.2 Avoid excessive personal centralization

If we rely too much on the central player in the soccer match, it will lead to the other players being passive and slack. The central player is more easily to be challenged by the opponents, leading to the negative situation.

In our model, the variance of contribution in the Huskies is 3 times over the average. **Such imbalance makes them weak when their leader player does not function well** .

In a social team, too much centralization of one person can also have a negative influence on the team. If other members of the team are overly dependent on the core member, the core member will overload and reduce his efficiency, and the other members will not devote themselves. What is more, if a person has too much power in the team, he will make himself profit as much as possible. This **demoralizes the team and harms the unity of the team**.

### 7.1.3 Stick to the Primary Goal

In the EM Model, we find that the Huskies shotting is the most important part of teamwork, and it is the ultimate goal of passing and defending. In a social team, we need to **identify the primary goal of the team and define specific implementation plans based on the primary goal**. If the team has a target, the team will struggle to find a breakthrough in the problem, which can greatly improve the efficiency.

#### 7.1.4 Keep Team Attributes Balanced

In the EM Model, we can see Huskies shotting and defending are almost equally important. It is impossible to become a brilliant team if the team only good at offense or defence.

In general, team efficiency will be greatly impacted if the team has obvious weakness. Each team member should also try to maintain balance.

### 7.2 Other Aspects of Team Network

In a network network, we usually focus on the outcome to evaluate team performance, which is not objective because **unsatisfying outcome does not mean unsuccessful teamwork**. We find some other aspects of team network to analyze team performance.

#### 7.2.1 Quality of Edges

We use edges to describe connections between members, but connection qualities are different. If we think about the **connecting qualities of edges** instead of just the number of edges, we can better describe **passing qualities or effective information transmission**. It describe the differences of exporting and receiving information among team members. Then we can further analyze individual connection skills and the flow of information throughout the team.

#### 7.2.2 Random Fluctuations of Each Node

Team work is always influenced by **external environment**, like different moods at work or different work environments. These external influence can be represented by random fluctuations of the nodes.

#### 7.2.3 Network Changes Over Time

Members of team is not always consistent. There may be **new nodes and new subnetworks arise over time**representing team personnel changes and network structure changes. It would be more effective and authentic if we can develop a dynamic network to analyze team performance.

## 8 Sensitivity Analysis

In the previous sections, we used all data to build passing network models to analyse the structure of the cooperation among players. In this part, we analyse the effect of **further data selection** on our model.

We find some passes do not precisely describe the offensive cooperation among players, such as stolen pass, bad passes, which lead to inoffensive plays. In real matches, good performance and offensive teamwork are usually performed by **continuous, consistent and offensive passing**. We call these passes effective passes. The previous network consider all the passes in the team, including ineffective passes. So we extract effective passes and build an effective connection network.

In professional soccer games, a consistent tactical execution is basically accomplished by the passing and movement of the players, such as attack through the middle, attempt through balls and steal. These significant tactics consist of players' continuous passing and movemont. So the network is developed considering the number of consecutive passes until the ball is lost.

## 8.1 Build the network

According to [4], the offensive play considers all the passes from the same offensive sequence without losing ball possession, who define each "unit of attack" (a soccer offensive play) starting at the moment a team gained ball possession until the ball was recovered by the opposing team. Furthermore, we take each player's holding time into consideration, for the reason that if a player possess a ball for a long time without losing the ball, we can also think the player does not connect with his teammates and execute the tactic.

We use the following algorithm to build the effective connection network.

**Algorithm 2:** extract effective passing data

**Input:** original passing data($p_o[n]$)
**Output:** network matrix $\mathbf{P} = [p_{ij}] \in \mathbb{R}^{n\times n}$ and effective passing data($p_e[n]$)

```
for k=1:n do
    if p_o[k].receiver = p_o[k+1].passer and p_o[k+1].time − p_o[k].time < 30s then
        copy p_o[k] into p_e;
        find p_o[k].passer and p_o[k].passer in P as i, j;
        p_ij = p_ij + 1;
output P, p_e[n] ;
```

First, we read a pass data and compare the current data with the next pass. If the receiver of the current pass is the passer of the next pass and two passes do not have long intervals, we think an effective connection starts. Then repeat the comparison step until a pass does not satisfy requirements of effective passing, and this pass is the end of an unit of effective connection. We keep these effective passes and reject other ineffective passes. After we traverse all passing data of the team, we find all units of effective connection and consistent passes.

Second, we define the corresponding graph $G_2 = (V, E)$where $V$ is the vertex and $E$ is the edge. According to the first model, the collection of vertexs is $V = \{v_{xx}|xx = g1, d1, d2, \cdots\}$. We say the number of players on the Huskies taking part in the game is $n$. For the convenience of expression, we use the index of each element to represent each vertex and player, namely $V = \{v_i|i = 1, 2, \cdots, n\}$,

Third, we use weighted adjacency matrix $\mathbf{P} = [p_{ij}] \in \mathbb{R}^{n\times n}$ to describe effective connection network. We define $p_{ij} = 0, (i = j)$and $p_{ij} =$ *the total number of effective passes from i to j*, $(i \neq j)$.

## 8.2 The Result and Analysis

We calculate the indicators again with this advanced network and find the main characteristic of the Huskies and their opponents keep unchanged. The only property that change considerably is the **contribution distribution**. The forwards gain more weights while the defenders less. Interestingly, the variance of contribution only drop slightly, from 3.08 to 2.89.

It is easy to understand such change, as we consider more offensive factors while choosing the data. To verify the stability of our result, we further rule out some data and find the result of

the Huskie remain stable while the opponents are more sensitive as their is only two matches for each opponent. If there is more data, they will be fine.

# 9 Strengths and Weaknesses

## 9.1 Strengths

- Detailed Explanation of the Motivations of Each Part of the Model.
- Comprehensive Analysis of the Team Features: we combine the result of our model and the knowledge of soccer to give reasonable descriptions of the strategy.
- Serious Sensitivity Analysis and Good Result: We describe how we seclect data with several criteria and havest good result. Explained in sensitivity analysis.

## 9.2 Weaknesses

- Our model requires lots of human based work while analysing the.
- The entropy method we used can not describe complicated relation between indicators and the final score.
- Some algorithm of the cooperative network is computationally intensive. Impractical for large network.

# 10 Conclusion

We build two network models and a entropy method weighing model in this paper to explore the teamwork structure of a soccer team.

After building the network, Se extract indicators from it and weigh them with entropy method. Then we test our theory with the data. We give advise and generalization after we verify our model.

In modeling, we found some factors in this model not easy to be quantified. Thus we describe some feature by the variable that suddenly hit us. We determine some expressions by observation of their beauty of form, also by trials and errors. Such exploration may not provide accurate result but still meaningful.

The real world presents all sorts of practical problems for us to think about and develop mathematical tools to solve. Our attempt, as a team, to gain insight into this problem provide a method to generalize discoverys while there is still much work to be done.

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# Appendices Appendix 1

## Code for Building Network

```
import xlrd
import numpy as np

pos = {'_F1': 0, '_F2': 1, '_F3': 2, '_F4': 3, '_F5': 4, '_F6': 5,
       '_D1': 6, '_D2': 7, '_D3': 8, '_D4': 9, '_D5': 10, '_D6': 11, '_D7': 12, '_D8': 13, '_D9': 14, '_D10': 15,
       '_M1': 16, '_M2': 17, '_M3': 18, '_M4': 19, '_M5': 20, '_M6': 21, '_M7': 22, '_M8': 23, '_M9': 24, '_M10': 25, '_M11': 26, '_M12': 27, '_M13': 28,
       '_G1': 29, '_G2': 30,
       0: '_F1', 1: '_F2', 2: '_F3', 3: '_F4', 4: '_F5', 5: '_F6',
       6: '_D1', 7: '_D2', 8: '_D3', 9: '_D4', 10: '_D5', 11: '_D6', 12: '_D7', 13: '_D8', 14: '_D9', 15: '_D10',
       16: '_M1', 17: '_M2', 18: '_M3', 19: '_M4', 20: '_M5', 21: '_M6', 22: '_M7', 23: '_M8', 24: '_M9', 25: '_M10', 26: '_M11', 27: '_M2', 28: '_M13',
       29: '_G1', 30: '_G2'}

def formal_pos(platerid):
    if platerid[-3]=='_':
        return platerid[-3:]
    else:
        return platerid[-4:]

class Graph:
    V = ['_F1', '_F2', '_F3', '_F4', '_F5', '_F6', '_D1', '_D2', '_D3', '_D4', '_D5', '_D6', '_D7', '_D8', '_D9',
         '_D10', '_M1', '_M2', '_M3', '_M4', '_M5', '_M6', '_M7', '_M8', '_M9', '_M10', '_M11', '_M12', '_M13', '_G1',
         '_G2']
    subV = np.zeros(31)
    E = np.zeros((31, 31))
    TeamID = ''
    MatchID = []

    def __init__(self, TeamID, MatchID):
        self.TeamID = TeamID
        for i in range(len(self.V)):
            self.V[i] += TeamID
        self.MatchID = MatchID

    def build(self):
        data = xlrd.open_workbook(r"data\passingevents.xlsx")
        table = data.sheets()[0]
        rows = table.nrows-1
        col_MatchID = [int(i) for i in table.col_values(0)[1:]]
        col_TeamID = table.col_values(1)[1:]
        col_OriginPlayerID = table.col_values(2)[1:]
        col_DestinationPlayerID = table.col_values(3)[1:]

        temp = np.zeros(31)
        for player in col_OriginPlayerID[:565]:
            if player[0]!='H':
                continue
            temp[pos[formal_pos(player)]] = 1
            # print(pos[formal_pos(player)])
        for player in col_DestinationPlayerID[:565]:
```

```
            if player[0]!='H':
                continue
            temp[pos[formal_pos(player)]] = 1
            # print(pos[formal_pos(player)])

        self.subV = np.argwhere(temp==0)

        for id in self.MatchID:
            for i in range(rows):
                if id == col_MatchID[i] and self.TeamID == col_TeamID[i]:
                    x = pos[formal_pos(col_OriginPlayerID[i])]
                    y = pos[formal_pos(col_DestinationPlayerID[i])]
                    self.E[x, y] += 1
        # print(self.E)

    def adjMatrix(self):
        return self.E

    def substi(self):
        return self.subV

    def reverse_adjM(self):
        E = self.E.copy()
        for i in range(31):
            for j in range(31):
                E[i, j] = 1 / E[i, j]
        return E

def build_graph(TeamID, MatchID):
    G = Graph(TeamID, MatchID)
    G.build()
    np.save('G.npy', G.adjMatrix())
    return G.adjMatrix()
```

# Appendix 2

## Code of Detect Motifs

```
import numpy as np
import networkx as nx
import matplotlib.pyplot as plt
from itertools import permutations
import time
import random


def calculateDegreeDistribution(G):
    NETWORK_SIZE = G.shape[0]
    averageDegree = 0.0
    identify = 0.0
    statistic = np.zeros((NETWORK_SIZE), dtype=float)
    in_degree = np.zeros((NETWORK_SIZE), dtype=int)
    out_degree = np.zeros((NETWORK_SIZE), dtype=int)
    for i in range(NETWORK_SIZE):
        for j in range(NETWORK_SIZE):
            if G[i, j] > 0:
                in_degree[j] += 1
                out_degree[i] += 1
    return in_degree, out_degree
    for i in range(NETWORK_SIZE):
```

```
        averageDegree += degree[i]
    averageDegree /= NETWORK_SIZE
    for i in range(NETWORK_SIZE):
        statistic[degree[i]] = statistic[degree[i]] + 1
    for i in range(NETWORK_SIZE):
        statistic[i] = statistic[i] / NETWORK_SIZE
        identify = identify + statistic[i]
    identify = int(identify)
    f = open('degree01.txt', 'w+')
    for i in range(NETWORK_SIZE):
        f.write(str(i))
        f.write(' ')
        s = str(statistic[i])
        f.write(str(s))
        f.write('\n')
    f.close()


def generate_graph(din, dout):
    D = nx.directed_configuration_model(din, dout)
    return D
    # nx.draw(D)
    # plt.show()

def find_motif(G):
    motif = np.zeros(13)
    n = G.shape[0]
    # n = 30
    # print(G)
    model = list(permutations([i for i in range(n)], 3))  # [(1, 2, 3), (1, 2, 4)...]
    for elem in model:
        # motif 1
        # print(elem)
        if G[elem[1], elem[0]]>0 and G[elem[1], elem[2]]>0:
            motif[0] += 1
        # motif 2
        if G[elem[1], elem[0]]>0 and G[elem[2], elem[1]]>0:
            motif[1] += 1
        # motif 3
        if G[elem[1], elem[0]]>0 and G[elem[1], elem[2]]>0 and G[elem[2], elem[1]]>0:
            motif[2] += 1
        # motif 4
        if G[elem[1], elem[0]]>0 and G[elem[2], elem[0]]>0:
            motif[3] += 1
        # motif 5
        if G[elem[1], elem[0]]>0 and G[elem[1], elem[2]]>0 and G[elem[2], elem[0]]>0:
            motif[4] += 1
        # motif 6
        if G[elem[1], elem[0]]>0 and G[elem[1], elem[2]]>0 and G[elem[2], elem[0]]>0 and G[elem[2], elem[1]]>0:
            motif[5] += 1
        # motif 7
        if G[elem[0], elem[1]]>0 and G[elem[1], elem[2]]>0 and G[elem[2], elem[1]]>0:
            motif[6] += 1
        # motif 8
        if G[elem[1], elem[0]]>0 and G[elem[0], elem[1]]>0 and G[elem[1], elem[2]]>0 and G[elem[2], elem[1]]>0:
```

```
            motif[7] += 1
        # motif 9
        if G[elem[0], elem[1]]>0 and G[elem[1], elem[2]]>0 and G[elem[2], elem[0]]>0:
            motif[8] += 1
        # motif 10
        if G[elem[0], elem[1]]>0 and G[elem[1], elem[0]]>0 and G[elem[1], elem[2]]>0 and G[elem[2], elem[0]]>0:
            motif[9] += 1
        # motif 11
        if G[elem[0], elem[1]]>0 and G[elem[1], elem[0]]>0 and G[elem[2], elem[1]]>0 and G[elem[2], elem[0]]>0:
            motif[10] += 1
        # motif 12
        if G[elem[0], elem[1]]>0 and G[elem[1], elem[0]]>0 and G[elem[2], elem[1]]>0 and G[elem[2], elem[0]]>0 and G[elem[1], elem[2]]>0:
            motif[11] += 1
        # motif 13
        if G[elem[0], elem[1]]>0 and G[elem[1], elem[0]]>0 and G[elem[2], elem[1]]>0 and G[elem[1], elem[2]]>0 and G[elem[0], elem[2]]>0 and G[elem[2], elem[0]]>0:
            motif[12] += 1
    return motif


def score(G):
    dis = calculateDegreeDistribution(G)
    # print(dis)
    din = dis[0][dis[0] > 0]
    dout = dis[1][dis[1] > 0]

    rand_G = generate_graph(din, dout)
    # print(calculateDegreeDistribution(nx.to_numpy_matrix(rand_G)))
    return find_motif(np.array(nx.adjacency_matrix(rand_G).todense()))
```

# Appendix 3

## Code of Extract Graph Feature

```
from buildgraph import *
import networkx as nx
import pandas as pd
import numpy as np
import csv


def degree(teamid):
    print(teamid)
    match = [i + 1 for i in range(38)]
    adjM = build_graph(teamid, match)
    de = []
    for i in range(31):
        t = 0
        for j in range(31):
            t += adjM[i, j]
        de.append(int(t))
    # graph = nx.from_numpy_matrix(adjM)
    # de = graph.degree()
    # print(de)

```

```
    averageDegree = 0
    NETWORK_SIZE = 30
    for i in range(NETWORK_SIZE):
        averageDegree += de[i]
    averageDegree /= NETWORK_SIZE
    # de = [int(i/19) for i in de]
    statistic = [0 for i in range(200)]
    for i in range(NETWORK_SIZE):
        print(de[i])
        statistic[de[i]] += 1
    return statistic
    print(statistic)
    t = nx.degree_hi
    stogram(graph)
    tt = []
    for i in t:
     if i != 0:
         tt.append(i)
    return np.var(tt)

def cluster(teamid):
    print(teamid)
    match = [i + 1 for i in range(38)]
    adjM = build_graph(teamid, match)
    graph = nx.from_numpy_matrix(adjM)
    return round(nx.average_clustering(graph), 4)
    # temp = nx.clustering(graph)
    # li = []
    # for i in range(31):
    #     li.append(round(temp[i], 4))
    # print(li)


motif_score = [95, 44, 64, 62, 128, 151, 44, 110, 93, 144, 193, 200, 219]
team_score = []

with open('score.csv')as f:
    f_csv = csv.reader(f)
    for row in f_csv:
        t = 0
        for i in range(13):
            t += float(row[i]) * motif_score[i]
        team_score.append(round(t, 4))

s = degree('Opponent4')
```

# Appendix 4

## Code of Shortest Path Adjacency Matrix

```
from buildgraph import *
import numpy as np
import networkx as nx
import csv


G = Graph('Huskies', [i + 1 for i in range(38)])
G.build()
re_G = nx.from_numpy_matrix(G.reverse_adjM())
print(nx.diameter(re_G))
```

```python

# print(nx.dijkstra_path(re_G, 29, 0))
# print(nx.dijkstra_path(re_G, 29, 18))
# print(nx.dijkstra_path(re_G, 18, 0))
# print(nx.dijkstra_path(re_G, 6, 10))
# floyd = nx.floyd_warshall_numpy(re_G)

for i in range(31):
    for j in range(31):
        floyd[i, j] = round(floyd[i, j], 2)

floyd = floyd.tolist()


headers = ['_F1', '_F2', '_F3', '_F4', '_F5', '_F6', '_D1', '_D2', '_D3', '_D4', '_D5', '_D6', '_D7', '_D8', '_D9',
           '_D10','_M1', '_M2', '_M3', '_M4', '_M5', '_M6', '_M7', '_M8', '_M9', '_M10', '_M11', '_M12', '_M13','_G1', '_G2']
rows = []
for i in range(31):
    rows.append([headers[i]] + floyd[i])
with open('floyd.csv', 'w', newline='')as f:
    f_csv = csv.writer(f)
    f_csv.writerow(headers)
    f_csv.writerows(rows)
```

# Appendix 5

## Code of Z-score Matrix

```python
from motif import *
from buildgraph import *
import numpy as np
import csv


def calc(teamid):
    m = 5
    motif_matrix = np.zeros((m + 1, 13))
    # motif  1  2  3 ... 13
    # real  ..
    # r_1      ..
    # r_2          ..
    # ..                ..
    # r_m
    # G = np.load('G.npy')
    G = build_graph('Opponent'+str(teamid), [i + 1 for i in range(38)])
    motif_matrix[0, :] = find_motif(G)

    for i in range(m):
        motif_matrix[i + 1, :] = score(G)

    Z_score = np.zeros(13)
    for i in range(13):
        N_real = motif_matrix[0, i]
        N_rand = sum(motif_matrix[1:, i]) / m
        sigma_rand = np.std(motif_matrix[1:, i], ddof=1)
        Z_score[i] = (N_real - N_rand) / sigma_rand

    temp = (Z_score - np.mean(Z_score)) / np.std(Z_score)
```

```
    t = [round(i, 2) for i in temp]
    print(t)
    arg = np.argsort(-Z_score) + 1
    print(arg)
    return arg


# headers = ['motif'+str(i+1) for i in range(13)]
rows = []
for i in range(19):
    print(i+1)
    rows.append(list(calc(i+1)))

with open('stat.csv', 'w', newline='')as f:
    f_csv = csv.writer(f)
    # f_csv.writerow(headers)
    f_csv.writerows(rows)
```

# Appendix 6

## Code of Entropy Method Model

```
function [s,w]=entropy(x,ind)

[n,m]=size(x);

for i=1:m
    if ind(i)==1
        X(:,i)=standardize(x(:,i),1,0.0002,0.999);
    else
        X(:,i)=standardize(x(:,i),2,0.0002,0.999);
    end
end

for i=1:n
    for j=1:m
        p(i,j)=X(i,j)/sum(X(:,j));
    end
end

k=1/log(n);
for j=1:m
    e(j)=-k*sum(p(:,j).*log(p(:,j)));
end
d=ones(1,m)-e;
w=d./sum(d);
s=100*w*X';

function y=standardize(x,type,ymin,ymax)
[n,m]=size(x);
y=zeros(n,m);
xmin=min(x);
xmax=max(x);
switch type
    case 1
        for j=1:m
            y(:,j)=(ymax-ymin)*(x(:,j)-xmin(j))/(xmax(j)-xmin(j))+ymin;
        end
    case 2
        for j=1:m
```

```
            y(:,j)=(ymax-ymin)*(xmax(j)-x(:,j))/(xmax(j)-xmin(j))+ymin;
        end
end


clc;clear;
M = csvread('data.csv');
Ind=[1 1 1 2];
[S,W]=entropy(M,Ind);
```